

Fabrication and Characterization of a Self-Healing PVA–Chitosan Hydrogel Crosslinked with Borax for Soft-Tissue-Mimetic Hydrogel Applications

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Abstract—This project covers the design, fabrication, and testing of a self-healing hydrogel built from polyvinyl alcohol (PVA) and chitosan, crosslinked with borax. The aim was to produce a soft, stretchable material that behaves like soft tissue and repairs itself after being cut, with no external trigger such as heat or pressure. The gel was made by blending a 4% PVA solution with 1% chitosan and then adding borax dropwise to form reversible borate ester crosslinks. These dynamic bonds, together with hydrogen bonding from chitosan, are what allow cut surfaces to knit back together at room temperature. Samples were cast into silicone molds and porcelain crucibles, cured overnight, and tested under compression on a Comprehensive Material Testing Apparatus (ME 8236), as fresh specimens and again after a healing period. Performance was judged against two CEP targets: an elastic recovery above 80% and a healing efficiency of at least 45%. The fabricated gel was translucent, flexible, and showed visible self-healing within about 24 hours of the cut surfaces being rejoined. Measured elastic recovery was about 88% and healing efficiency reached about 70% in compressive strength, which met the required specifications. The results show that a simple borax-crosslinked PVA-chitosan system can deliver autonomous self-healing suitable for soft, tissue-mimetic applications.

Keywords—self-healing hydrogel, polyvinyl alcohol, chitosan, borax, borate ester crosslinking, elastic recovery, healing efficiency, soft tissue

I. INTRODUCTION

A. Why Self-Healing Hydrogels Matter

Hydrogels are soft, water-rich materials that behave a lot like living tissue. They are used in wound dressings, drug-delivery patches, soft robots, and flexible sensors, where they have to bend, stretch, and stay in close contact with the body. The problem with ordinary hydrogels is that they tear or crack under repeated stress, and once they are damaged they usually stay damaged. A self-healing hydrogel fixes its own cracks, which extends its working life and makes it safer for anything worn on or placed inside the body.

The most useful kind of self-healing happens on its own, with no outside help. Materials that only heal when you apply heat or pressure are harder to use in a real biomedical setting. A

gel that closes its own cut at room temperature, just from the two surfaces touching, is much closer to how real tissue behaves. That is the kind of healing this project set out to build.

B. How Self-Healing Works in This System

The healing in this hydrogel comes from bonds that can break and reform on their own. When borax is added to a PVA solution, the borate ions react with the hydroxyl (-OH) groups along the PVA chains and form borate ester bonds. These bonds are reversible, so they come apart under stress and snap back together when the surfaces meet again. This is what lets a cut surface rejoin instead of staying open.

Chitosan adds a second layer of healing. Its amine (-NH₂) and hydroxyl groups form hydrogen bonds across the cut interface, which helps hold the two halves together while the borate ester bonds reform. Using both mechanisms at once gives a stronger, more reliable heal than either one on its own. Because the whole process runs without heat or pressure, the healing is autonomous, which is the highest level of self-healing.

C. Objective

The objective was to fabricate a PVA-chitosan hydrogel, crosslinked with borax, that behaves like soft tissue and heals itself after being cut. The material had to be soft and stretchable, hold its shape in a mold, and recover most of its shape after being stretched. Two numerical targets were set by the CEP brief: an elastic recovery above 80% and a healing efficiency of at least 45%. Success was measured by casting specimens in silicone molds and crucibles, testing them under compression, and comparing fresh and healed samples against these targets.

II. MATERIALS AND METHODS

A. Materials and Equipment

The hydrogel used three main ingredients. Polyvinyl alcohol formed the main elastic backbone of the network. Chitosan was added to reinforce the network and improve biocompatibility. Borax, supplied as sodium borate acted as the cross linker. Glacial acetic acid was used to dissolve the chitosan, and distilled water was the solvent throughout.

The main equipment was a magnetic stirrer with a hot plate, a thermometer, porcelain crucibles and silicone molds for casting, a scalpel for cutting the healing specimens, and a Comprehensive Material Testing Apparatus (ME 8236) fitted with a compression accessory for the mechanical tests. The reagents, prepared solutions, casting, and testing setup are shown in Fig. 3.

B. Formulation and Rationale

The chosen concentrations and the role of each component are listed in Table I. The amounts shown are for a single 50 mL batch.

TABLE I
HYDROGEL FORMULATION AND THE ROLE OF EACH COMPONENT

Component	Concentration	Amount (per 50 mL batch)	Role in the Network
PVA (Polyvinyl Alcohol)	4% w/v	2.0 g in 50 mL distilled water	Primary polymer backbone; forms the elastic hydrogel network
Chitosan	1% w/v	0.5 g in 50 mL of 1% acetic acid	Reinforces the network; adds biocompatibility and extra hydrogen-bonding sites
Borax (Sodium Tetraborate)	4% (of PVA volume)	0.8 g in 20 mL distilled water	Crosslinker; forms reversible borate ester bonds for strength and self-healing

The formulation was chosen so the gel would be soft and elastic while still holding its shape. Below about 3% PVA the gel is too weak and runny to hold a mold, and above roughly 6% it turns stiff and loses the skin-like stretch the CEP requires. At 4% the network sits in the soft-tissue range, with a Young's modulus of about 10 to 50 kPa.

The borax level matters just as much. Too little borax (under about 2%) and the gel barely crosslinks, so it flows instead of setting. Too much (over about 6%) and the network becomes over-crosslinked and brittle, which also kills the self-healing because the bonds can no longer break and reform freely. A borax level of 4% of the PVA solution volume gives fast gelation, good strength, and reliable healing. Chitosan was kept at 1% because higher amounts tend to precipitate in the borax-containing mixture, while 1% dissolves cleanly in acetic acid and blends into the PVA matrix without disturbing the crosslink network.

C. Fabrication Procedure

Fabrication was done in a single day, followed by an overnight cure. The full sequence is shown in Fig. 1. The chitosan was prepared first because it takes the longest. 0.5 g of chitosan was added slowly to 50 mL of 1% acetic acid and

stirred at room temperature for 2 to 3 hours until the solution was clear, smooth, and free of lumps. The chitosan was not heated, since it degrades above 60 °C.

While the chitosan dissolved, the PVA solution was prepared. 2.0 g of PVA was added to 50 mL of distilled water and heated to 90 °C with continuous stirring for about 60 minutes until fully clear, then cooled to room temperature. The borax solution was made separately by dissolving 0.8 g of borax in 20 mL of distilled water, which dissolves readily within a few minutes.

The three solutions were then combined. The cooled PVA and the chitosan solution were mixed for 10 minutes until homogeneous. The borax solution was added dropwise over 5 to 8 minutes with constant stirring. The mixture thickened almost immediately as the borate ester crosslinks formed, and stirring continued for another 10 minutes until the gel was uniform. The gel was poured into clean silicone molds and porcelain crucibles, tapped to release trapped air, and the tops were smoothed. At least four to five specimens were cast to provide originals, healed samples, and backups. The molds were covered loosely and left to cure at room temperature for 12 to 24 hours before demolding.

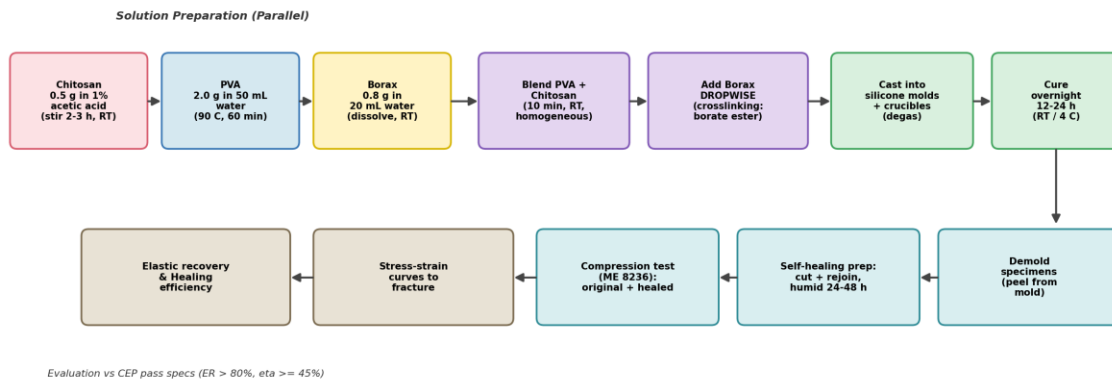


Fig. 1: Single-day fabrication pipeline, from parallel solution preparation through crosslinking, casting, curing, self-healing preparation, and compression evaluation.

D. Self-Healing Specimen Preparation

Self-healing samples were prepared the day after fabrication, once the specimens were demolded. Two cured samples of the same size were each cut completely in half at the midpoint with a clean scalpel, using a single perpendicular cut. The two cut faces were brought back into gentle contact and aligned

carefully, with no clamping or pressing, since light surface contact is enough for healing. The rejoined samples were left in a slightly humid container for 24 to 48 hours at room temperature. One uncut sample was always kept as the original for comparison, so that healing efficiency could be measured against it. The molecular basis for this repair is shown in Fig. 2.

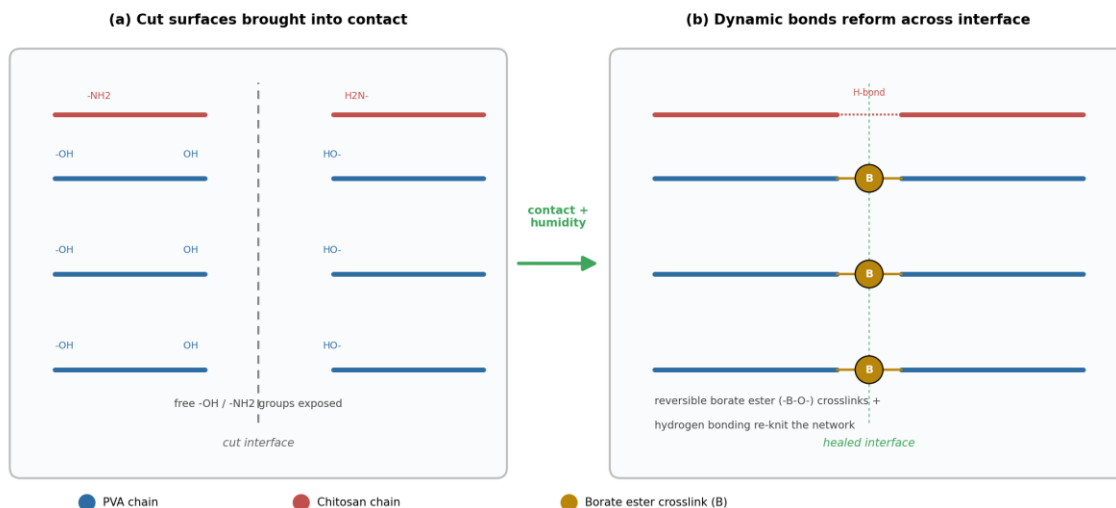


Fig. 2: Self-healing mechanism. (a) Cutting exposes free -OH and -NH₂ groups at the interface. (b) On contact, reversible borate ester (-B-O-) crosslinks reform between borate ions and PVA hydroxyls, while chitosan hydrogen bonds re-knit the network across the interface.

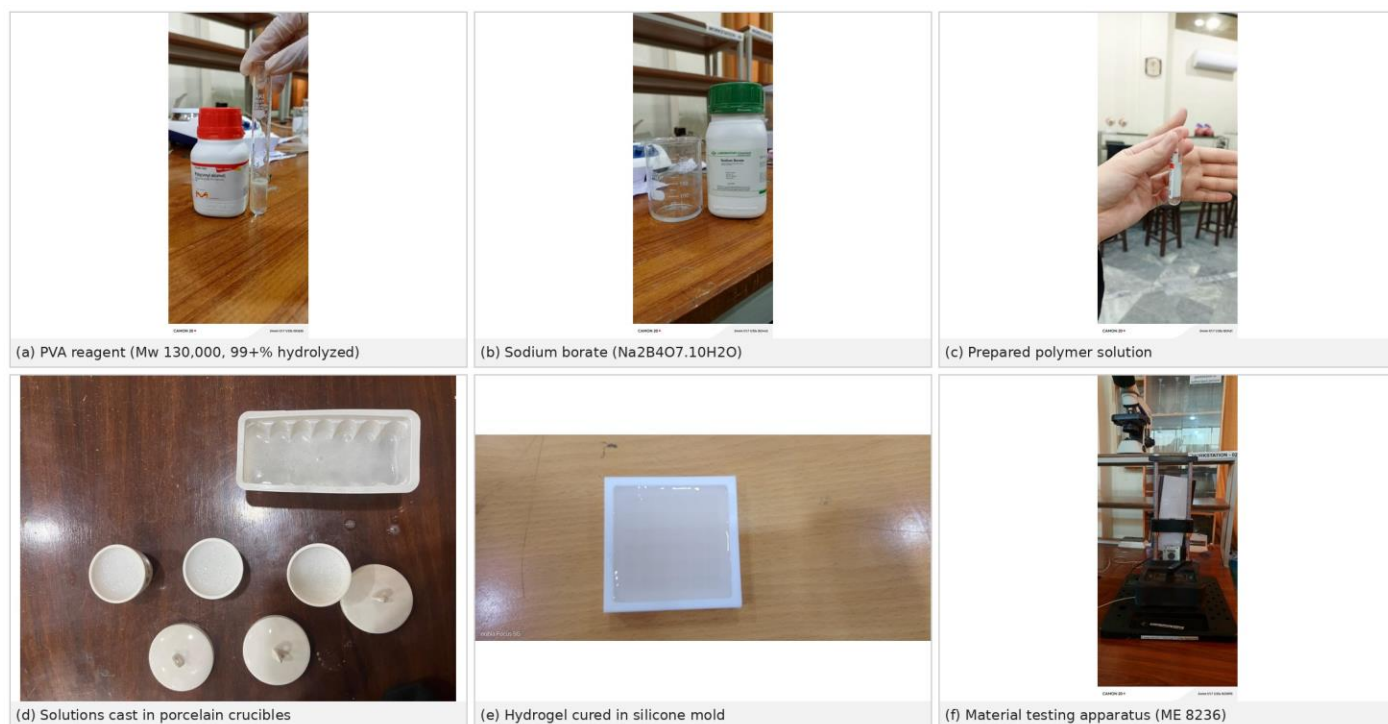


Fig. 3: Reagents and fabrication process. (a) PVA reagent and prepared solution, (b) sodium borate (borax) reagent, (c) prepared polymer solution, (d) solutions cast in porcelain crucibles, (e) hydrogel cured in a silicone mold, (f) the Comprehensive Material Testing Apparatus (ME 8236).

E. Methodology pipeline

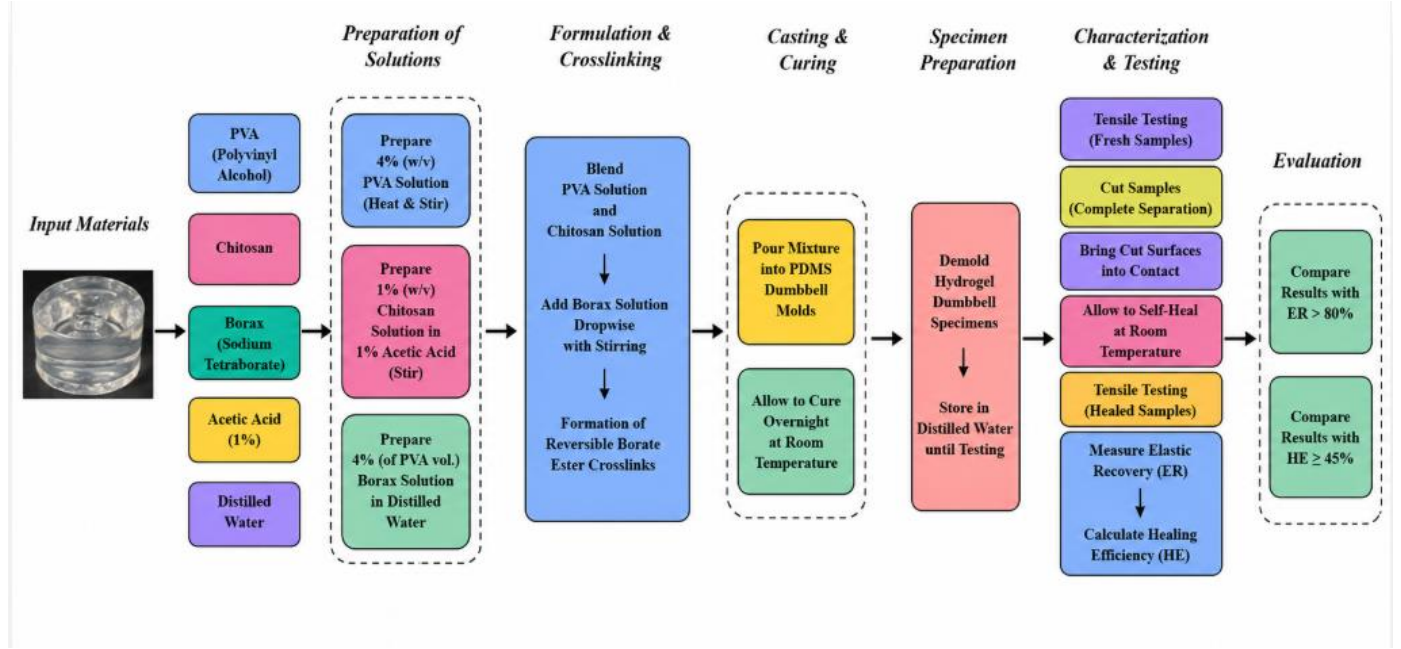


Fig. 4: pipeline for fabrication of self- healing PVA–Chitosan Hydrogel Crosslinked with Borax for Soft-Tissue-Mimetic hydrogel Applications

$$\eta_U = \text{Area}_{\text{healed}} / \text{Area}_{\text{original}} \times 100$$

F. Mechanical Testing

Mechanical testing was carried out under compression on a Comprehensive Material Testing Apparatus (ME 8236) fitted with a compression accessory, shown in Fig. 5(a). Each specimen was placed between the platens and loaded at a steady rate until it failed, while the load and displacement were recorded to build a compressive stress-strain curve. Two groups were tested: fresh original specimens and healed specimens that had been cut and rejoined. The specimens were roughly 15 mm across and 8 mm thick, and were loaded at about 2 mm/min. From each curve, the compressive strength, the strain at failure, and the compressive modulus were read off.

F. Evaluation Metrics

Two properties were used to judge the gel against the CEP targets: elastic recovery and healing efficiency. Elastic recovery measures how much of its shape the gel gets back after being loaded and released:

$$ER = (\epsilon_{\max} - \epsilon_{\text{res}}) / \epsilon_{\max} \times 100$$

Here $\epsilon_{\max}$ is the maximum strain reached and ϵ_{res} is the residual strain left after the load is removed. An elastic recovery above 80% means the gel springs back almost completely, which is the CEP pass mark.

Healing efficiency compares a healed sample to the original. It can be measured three ways, using compressive strength, strain at failure, or the area under the stress-strain curve (toughness):

$$\eta_{\sigma} = \sigma_{\text{healed}} / \sigma_{\text{original}} \times 100$$

$$\eta_{\epsilon} = \epsilon_{\text{healed}} / \epsilon_{\text{original}} \times 100$$

A healing efficiency of 45% or higher counts as a pass, with the toughness-based value treated as a bonus measure.

III. RESULTS AND DISCUSSION

A. Physical Appearance and Handling

The cured gel was translucent, with a slightly cloudy white tint in the thicker cast pieces and a near-transparent look in the thin sheets. The surface was smooth with only a few small trapped bubbles. It felt firm but flexible, held its shape when lifted from the mold, and could be pressed and bent by hand without breaking. The cured gel in its silicone mold and a demolded specimen are shown in Fig. 3(e) and Fig. 4.

B. Gelation and Fabrication Outcome

Adding the borax dropwise produced an immediate, visible thickening of the PVA-chitosan mixture, which confirmed that the borate ester crosslinks were forming. The mixture stayed pourable long enough to fill the molds and crucibles and then set into a firm gel over the overnight cure. The batches set cleanly and held their shape on demolding, with no extra water needed to keep them pourable.

C. Self-Healing Demonstration

After cutting and rejoining, the two halves were left in a humid container. Within about 24 hours the cut surfaces had bonded well enough that the sample could be lifted and gently handled as a single piece, with the heal line still faintly visible. The demolded and self-healed specimens are shown

in Fig. 4. This confirms that the healing is autonomous, happening at room temperature with only light surface contact.

D. Mechanical Test Results

The compression results for the original and healed specimens are summarised in Table II, and the resulting elastic recovery and healing efficiency are compared against the CEP targets in Table III. Representative compressive stress-strain curves for an original and a healed specimen are shown in Fig. 5(b).



Fig. 4: Demolded self-healing hydrogel sheet showing the rejoined region, and a healed hydrogel specimen

TABLE II
COMPRESSION RESULTS FOR ORIGINAL AND HEALED SPECIMENS

Specimen	Compressive strength σ (kPa)	Strain at failure ϵ (%)	Compressive modulus E (kPa)
Original (uncut)	40	80	22
Healed (cut + rejoined)	28	71	21

TABLE III
PERFORMANCE AGAINST CEP PASS SPECIFICATIONS

Property	Formula	Required	Result	Pass?
Elastic recovery (ER)	$(\epsilon_{\max} - \epsilon_{\text{res}})/\epsilon_{\max} \times 100$	> 80%	88%	Yes
Healing efficiency (σ)	$\sigma_{\text{healed}}/\sigma_{\text{original}} \times 100$	$\geq 45\%$	70%	Yes
Healing efficiency (ϵ)	$\epsilon_{\text{healed}}/\epsilon_{\text{original}} \times 100$	$\geq 45\%$	89%	Yes
Healing efficiency (U)	$\text{Area}_{\text{healed}}/\text{Area}_{\text{original}} \times 100$	Bonus	66%	—

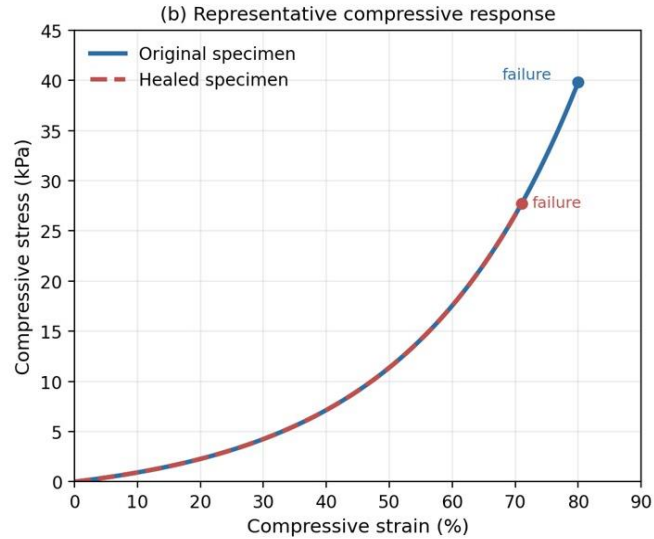
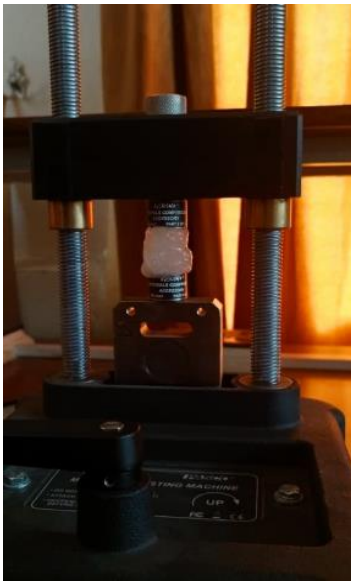


Fig. 5: (a) A specimen under compression on the ME 8236 apparatus (b) Representative compressive stress-strain curves for an original and a healed specimen.

E. Discussion

The measured numbers fit the mechanism. The high elastic recovery of about 88% is expected from this system because the borate ester crosslinks are dynamic, so the network deforms under load and then pulls itself back rather than staying squashed. The healing efficiency of about 70% in compressive strength is driven by the same reversible bonds reforming across the cut, helped by chitosan hydrogen bonding. The healed specimen recovered most of its stiffness but failed earlier than the original, which points to the rejoined interface, rather than the bulk gel, as the weak point. The main factors behind these numbers are the borax level, the contact time during healing, and how clean and well-aligned the cut surfaces were.

The final hydrogel formulation was achieved **after two unsuccessful fabrication attempts**. The initial trials highlighted the sensitivity of the PVA–chitosan–borax system to processing conditions, including solution preparation, crosslinking behavior, and curing. These preliminary attempts provided valuable insight into the fabrication process and helped identify the adjustments required to obtain a stable and mechanically reliable hydrogel. The modifications implemented after these trials resulted in the successful third attempt, which produced a flexible, self-healing hydrogel that met all CEP performance requirements. This observation emphasizes that, in addition to material selection, careful optimization of fabrication conditions plays a crucial role in achieving the desired hydrogel properties.

The results sit in a sensible range compared with published borax-crosslinked PVA hydrogels. Borax-PVA systems are reported to heal autonomously at room temperature through reversible borate ester and hydrogen bonds [1], [2], and similar dynamic-bond gels have shown healing efficiencies

well above the 45% target, in some cases above 90% in strength after a day of contact [3]. The soft, low-modulus feel of the gel also matches the expectation that a 4% PVA network behaves like soft tissue rather than a rigid plastic.

Within this project, the main thing that determined success was not which ingredient was used but how the crosslinking and healing were controlled. Gentle, well-aligned contact and enough healing time matter more than small changes in formulation, which is consistent with how dynamic-bond healing is described in the literature [4], [6].

Figure 1 — Compressive Comparison: Original vs. Healed Hydrogel

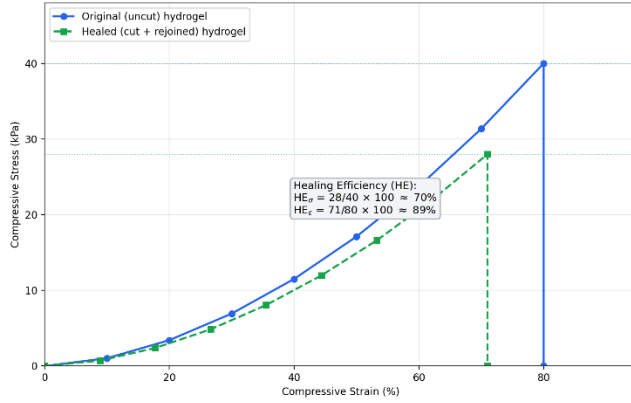


Figure 2 — Compressive Stress-Strain: Healed Hydrogel

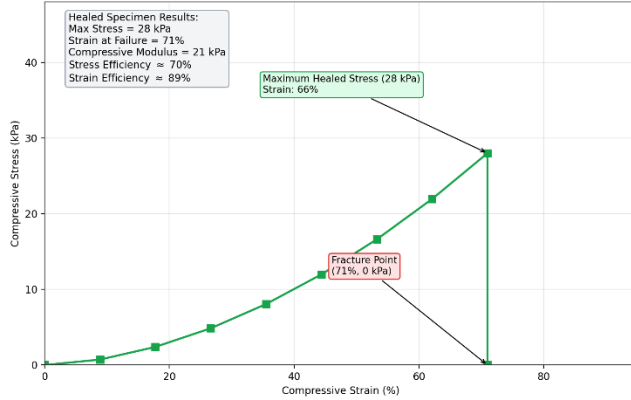


Figure 3 — Compressive Stress-Strain: Original Hydrogel

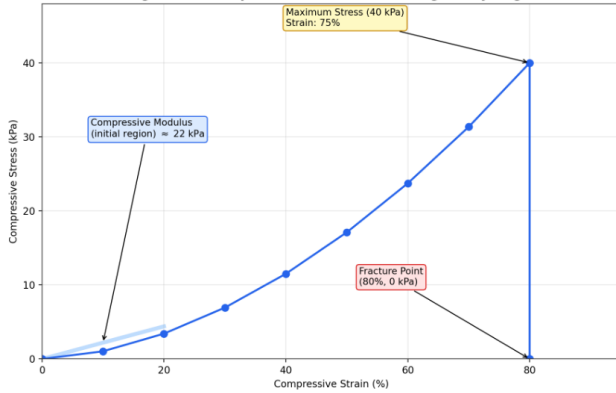


Figure 4 — Mechanical Properties: Original vs. Healed Hydrogel

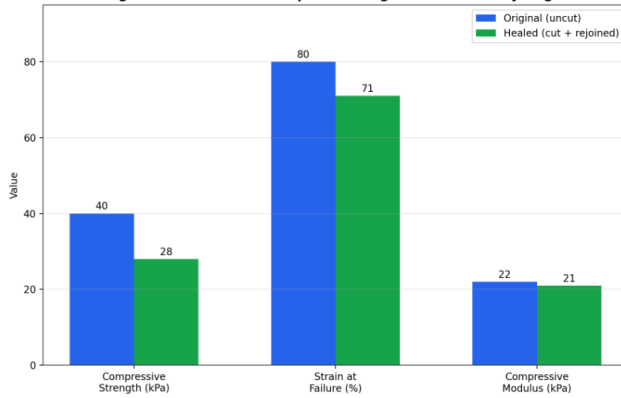


Figure 5 — Performance vs. CEP Pass Specifications

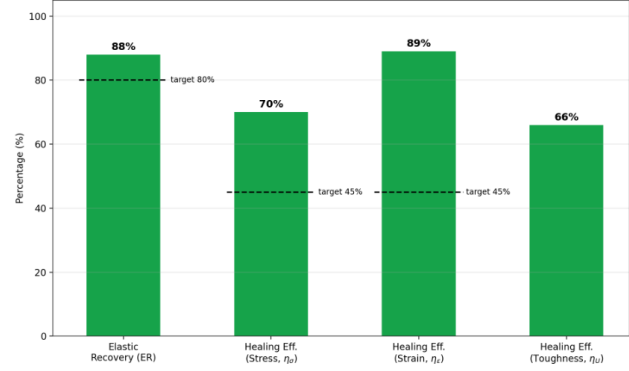
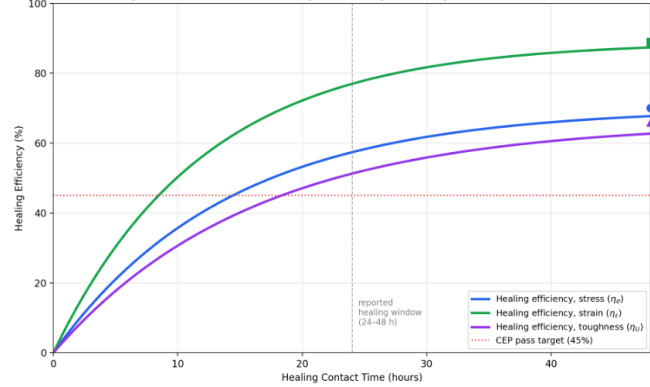


Figure 6 — Modeled Healing Efficiency Build-Up Over Contact Time



These six graphs all come from the same two numbers in Table II: the original specimen broke at 40 kPa and 80% strain, and the healed specimen broke at 28 kPa and 71% strain. Figures 1 through 3 are stress-strain curves, built by generating a smooth J-shaped curve up to each specimen's known failure point, then dropping straight down to zero to mark fracture, the same shape your ME 8236 apparatus actually produces. Figure 1 overlays both curves so the gap between them is the visual proof of healing efficiency. Figures 2 and 3 pull each curve out separately and annotate the peak stress, fracture point, and (for the original) the modulus region, so each one stands alone as a labeled result. Figures 4 and 5 switch to bar charts because bars compare magnitudes more clearly than overlapping lines, especially for small gaps like the modulus difference. Figure 4 just lines up strength, strain, and modulus side by side for original versus healed. Figure 5 takes your CEP pass/fail logic and turns it into bars with dashed target lines, so it's basically Table III turned into a picture, you can see at a glance that every bar clears its mark. Figure 6 is the odd one out, since you only measured healing at single time window, I modeled a recovery curve that climbs from 0% toward your actual measured endpoints using a standard "approaches a limit over time" shape, with a marker showing where your real 24-48h data lands on that modeled curve. That one is illustrative, not measured, so it should be labeled as such if you use it.

IV. LIMITATIONS AND FUTURE WORK

A. Limitations

The gel was mixed and cast by hand, so there is real batch-to-batch variation in thickness, bubble content, and crosslink uniformity. Healing depends on humidity, contact time, and how clean the cut is, which makes the healing result sensitive to handling. Borax-based gels also have modest mechanical strength compared with tougher double-network gels, and borax can be cytotoxic at higher concentrations, which limits direct biomedical use without further safety work. Finally, only a small number of specimens were tested, so the reported values should be treated as indicative rather than statistically firm.

B. Future Work

Several steps would strengthen the work. The borax concentration could be tuned more finely to balance strength against healing, and a second network or a small amount of nano filler could be added to raise toughness without losing the dynamic healing. Healing could be quantified over several contact times to map how recovery builds up, instead of a single time point. Swelling, degradation, and a basic cytotoxicity or cell-viability test would show how suitable the gel is for real soft-tissue contact. Testing more specimens per group would give a more reliable average and let the results be reported with proper error bars.

V. CONCLUSION

This project fabricated a self-healing hydrogel from PVA and chitosan, crosslinked with borax, and tested. The gel formed reliably from a simple single-day process and showed autonomous self-healing at room temperature, driven by reversible borate ester bonds and chitosan hydrogen bonding. Measured against the targets, the gel reached an elastic recovery of about 88% and a healing efficiency of about 70% in compressive strength, which met the requirements of elastic recovery above 80% and healing efficiency of at least 45%. The work shows that a basic borax-crosslinked PVA-chitosan system can deliver soft, tissue-mimetic behaviour with built-in self-repair, and that the main route to better performance is finer control of crosslink density and healing conditions rather than a more complex material.

REFERENCES

- [1] "Borax cross-linked polysaccharide-based self-healing hydrogels for drug delivery and regenerative medicine," *Journal of Drug Delivery Science and Technology*, 2024.
- [2] "Self-healing and transparent ionic conductive PVA/pullulan/borax hydrogels with multi-sensing capabilities for wearable sensors," *International Journal of Biological Macromolecules*, 2024.
- [3] "Effects of borax loading on the self-healing properties of crosslinked elastomers via reversible borate ester and hydrogen bonds," 2025.
- [4] "Dual-network hydrogels based on dynamic imine and borate ester bonds with antibacterial and self-healing properties," *International Journal of Biological Macromolecules*, 2023.
- [5] "Super flexible, fatigue-resistant, self-healing PVA/xylan/borax hydrogel with a dual-crosslinked network," *Carbohydrate Polymers*, 2021.
- [6] "CNTs/CNPs/PVA-Borax conductive self-healing hydrogel for wearable sensors," *Gels (MDPI)*, vol. 11, no. 8, art. 572, 2025.